KDGraph: A Keypoint Detection Method for Road Graph Extraction From Remote Sensing Images

Wei He, *Senior Member, IEEE*, Ruoyao Xue, Fangxiao Lu, Jinjun Xu, and Hongyan Zhang, *Senior Member, IEEE*

***Abstract*— Road graph extraction from remote sensing images is essential in navigation and urban planning. However, shadows and occlusions in these images frequently disrupt the continuity of road representations, resulting in fragmented and poorly connected road graphs extracted by existing methods. To overcome these challenges, we introduce KDGraph, a novel keypoint detection method for road graph extraction from remote sensing images. Specifically, keypoints are defined as vertices situated at the endpoints, corners, or junctions of roads, characterized by their positions and multiple directional attributes. In addition, we develop a greedy parsing algorithm to connect these keypoints and construct the road graph based on their directional information. The key innovation of our method lies in reformulating road mask extraction as a keypoint detection task. By positioning keypoints in areas less affected by shadows and occlusions, KDGraph effectively reduces their impact on road graph connectivity. Extensive experiments are conducted using the SpaceNet3 dataset and a newly constructed shadow-occluded road graph extraction (SoR) dataset, which includes road segments from 15 cities with varying degrees of shadows and occlusions. A patch expansion strategy is introduced for large-scale inference on the SoR dataset. Results show that KDGraph outperforms all comparative methods in both generalization and the capability of handling shadows and occlusions. Code and SoR dataset available at: https://github.com/ruoyxue/KDGraph.**

***Index Terms*— Deep learning, keypoint detection, remote sensing, road graph extraction.**

I. Introduction

ROAD graph extraction from remote sensing images is a critical component in both navigation systems [1], [2] and urban planning applications [3], [4]. A road graph is a vectorized representation of the topology of the road, consisting of vertices and edges that precisely delineate the structural layout of the road network. In most conventional approaches, the vertices are uniformly distributed along roadways, with edges linking the adjacent vertices to form the corresponding road segments [5], [6]. Historically, manual road graph construction was labor intensive, costly, and inefficient. In recent years, considerable progress has been made in automatic road graph extraction from remote sensing images [7], [8], [9]. These methods are generally classified into two main categories: edge-first methods [10], [11], [12] and vertex-first methods [13], [14], [15], [16].

Edge-first methods [17] focus on extracting road vertices using road masks. These approaches typically involve segmentation techniques to derive road centerlines, which are subsequently simplified to identify road vertices. Historically, these methods were based on manually designed topological and geometric features for the extraction of road masks [18], [19], [20]. However, with the advancement of deep learning [21], convolutional neural networks (CNNs) have increasingly been used to the automate extraction of road masks [22], [23], [24], [25], [26], [27], [28], [29]. In particular, HRNet [30] introduced subnetworks that operate at multiple resolutions, connected in parallel to preserve multiscale information throughout the encoding process. DDUNet [31] targeted narrow roads by incorporating a dilated convolution attention module, which captures multiscale features and expands the receptive field to improve road connectivity. GAMSNet [32] employed a global awareness operation to capture spatial context dependencies and channel-wise information. CoANet [33] developed a strip convolution module that aligns with road shapes to improve the extraction of features. MSSDMPA [34] introduced dynamic attention map-guided index pooling and dynamic attention map-guided spatial and channel attention mechanisms for precise road extraction. Despite these advancements, as shown in Fig. 1, edge-first methods are frequently hindered by incomplete intermediate road masks, resulting in fragmented road graphs in the presence of shadows and occlusions.

To mitigate the influence of incomplete road masks, vertex-first methods directly extract road vertices before generating the associated edges. The pioneering approach, RoadTracer [5] employed an iterative strategy, starting with an initial vertex and predicting the direction to the adjacent vertices using a classification model. The agent then moved a fixed distance along this direction to identify the new vertex, iterating until the entire road graph was mapped. On the basis of RoadTracer, VecRoad [13] improved the approach by using heatmap predictions to permit variable step lengths. RNGDet [14] incorporated transformers [35] to predict multiple adjacent vertices simultaneously, improving the handling of complex road junctions. Nevertheless, iterative methods often faced challenges of prolonged inference times and error

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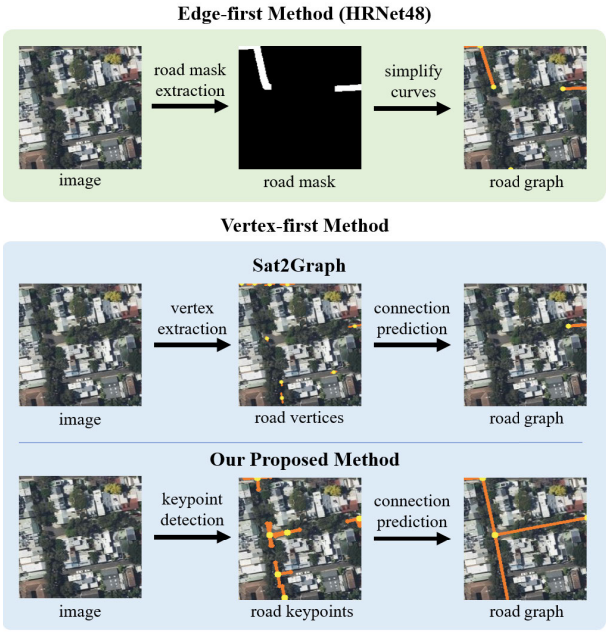


Fig. 1. Comparison of road graph extraction methods in tree-occluded areas. The proposed method exhibits superior robustness in extracting road graphs under occlusion, compared to the edge-first method (HRNet48) and the vertex-first method (Sat2Graph).

accumulation. In contrast, Sat2Graph [6] adopted an end-to-end approach for vertex extraction and developed a decoding algorithm to detect road edges. However, it generated ground truth vertices with even spacing from the segmentation map, which was physically meaningless and increased ambiguity in vertex prediction. As shown in Fig. 1, Sat2Graph produces fragmented road graphs in shaded or occluded areas due to missing vertices.

Existing methods are significantly affected by shadows and occlusions in remote sensing images. Observations indicate that shadows and occlusions primarily affect edge features of the roadway rather than endpoints, corners, or junctions. Constructing the road graph using prior knowledge of these keypoints is theoretically feasible and can alleviate the adverse effects of shadows and occlusions. However, connecting keypoints presents challenges due to uncertainties in road length, complex junction directions, and diverse keypoint connections. Existing vertex-first methods either employ an iterative strategy [5], [13] that generates edges iteratively or an evenly spaced strategy [6] that simplifies connections by placing vertices close together. Both strategies are notably affected by shadows and occlusions.

In this study, we define the keypoints as meaningful vertices located at the endpoints, corners, or junctions along the roadway and propose a novel keypoint detection method for road graph extraction from remote sensing images, termed KDGraph. The flowchart of KDGraph is illustrated in Fig. 2. In Phase I, we introduce a network to directly predict the road keypoint features. In Phase II, the trained model is used to detect road keypoints, and a greedy parsing algorithm is applied to generate edges. By reformulating road graph extraction as a keypoint detection task and positioning keypoints in less affected areas, KDGraph improves robustness against shadows and occlusions. Experiments are performed on the SpaceNet3 dataset [37] and the SoR dataset developed for this study. To facilitate large-scale road graph inference on the SoR dataset, a patch expansion strategy is introduced. Results demonstrate the superiority of KDGraph in both generalization and the capability of handling shadows and occlusions compared to existing methods. The main contributions of this study are summarized as follows.

1) We propose KDGraph that reformulates road graph extraction as a keypoint detection task and strategically positions keypoints in less affected areas, thereby mitigating the adverse effects of shadows and occlusions.
2) We propose a greedy parsing algorithm for keypoint connection that addresses challenges such as road length uncertainty, complex junction directions, and diverse connection possibilities.
3) To comprehensively assess the effectiveness of KDGraph, we construct the SoR dataset, and introduce a patch expansion strategy for large-scale inference. Extensive experiments show that KDGraph excels in both generalization and handling shadows and occlusions compared to other road graph extraction methods.

The rest of this article is organized as follows. Section II presents the ground truth generation of keypoint feature maps, two phases of KDGraph, and the patch expansion strategy. Section III introduces the experiments, including datasets, implementation details, benchmark evaluation metrics and methods, and extensive experiment results. The conclusion is presented in Section IV.

II. Proposed Method

The core concept of this study is to detect road keypoints located at endpoints, corners, and junctions using feature maps and then connect them to generate the road graph. The proposed KDGraph utilizes a network consisting of a shared encoder and two independent decoders to predict keypoint feature maps and a greedy parsing algorithm for keypoint connection. As illustrated in Fig. 2, the KDGraph framework consists of two phases: Phase I involves training the model by using ground truth keypoint feature maps, while Phase II focuses on connecting the keypoints based on directional information.

A. Ground Truth Generation of Keypoint Feature Maps

Road keypoints are defined as critical locations along the road centerline, representing significant geometric and topological characteristics.

As depicted in Fig. 3, junctions are located where three or more road segments converge, reflecting essential connectivity within the road graph. These junctions are connected to more than two neighboring keypoints. Corners, representing significant changes in road direction, are identified using the Ramer–Douglas–Peucker algorithm [38] and typically have two neighboring keypoints. Endpoints, which mark the start or end of a road segment, are connected to a single neighboring

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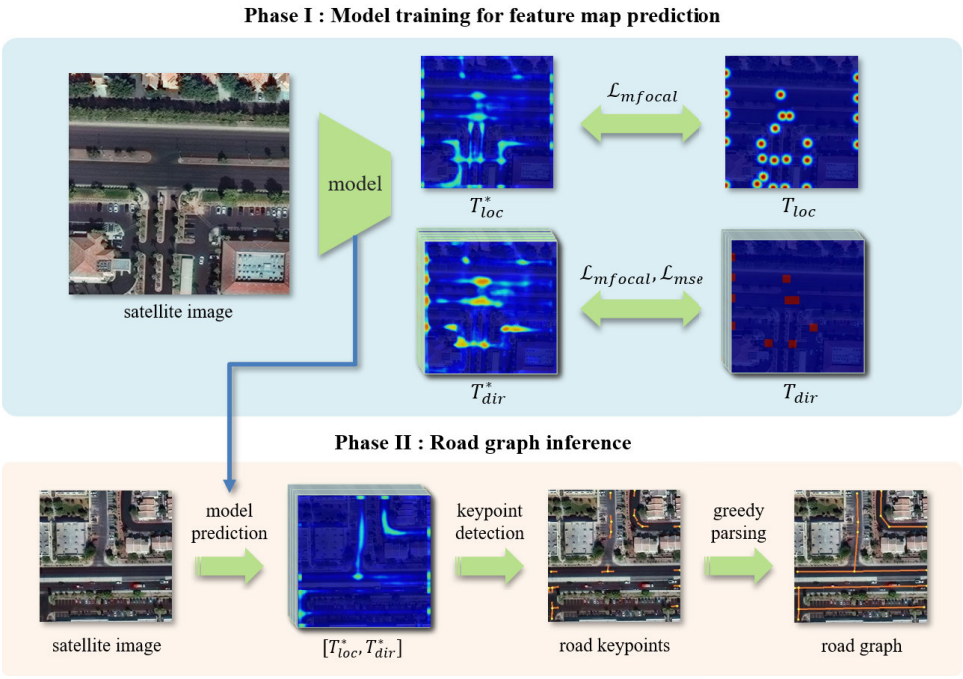


Fig. 2. Overall pipeline of the proposed method. T_{loc} denotes the ground truth location feature map, while T_{dir} denotes the direction feature maps. T^*_{loc} and T^*_{dir} are their predicted counterparts. $\mathcal{L}_{\text{mfocal}}$ denotes the modified focal loss [36] and $\mathcal{L}_{\text{mse}}$ denotes the mean squared error loss.

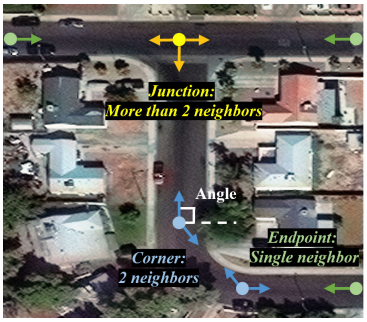


Fig. 3. Diagram of keypoint classification: junction, corner, and endpoint. Each keypoint comprises a location and angles directed to neighboring keypoints.

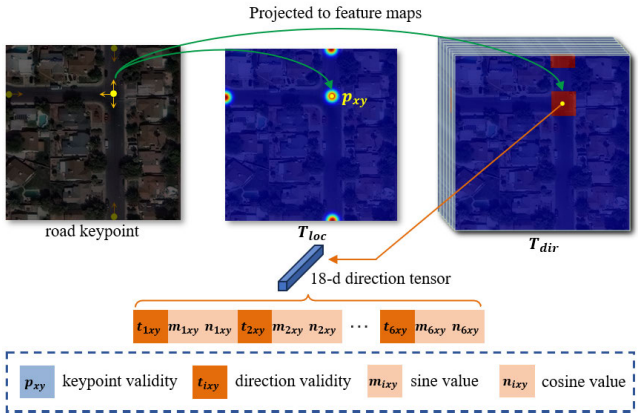


Fig. 4. Keypoint feature maps generated from road keypoints (with $D_{\max} = 6$).

keypoint. These keypoints provide a compact and precise representation of the road structure. Each keypoint is defined by its location and directions, with directions represented by the sine and cosine values of the angle toward a neighboring keypoint.

Typically, existing road graph datasets are used to generate the ground truth for keypoint feature maps. However, road vertices in the original datasets are manually annotated along the road and may not accurately reflect keypoint locations. To address this issue, we apply the Ramer–Douglas–Peucker algorithm to simplify road vertices into endpoints, corners, or junctions. The ground truth of keypoint feature maps is derived from these simplified keypoints. To avoid overfitting, we encode the location and directions of a keypoint as related features with Gaussian and identity distributions in the feature maps, respectively, rather than directly marking individual pixels. Next, we describe the process of generating the ground truth location and direction feature maps for keypoints.

For a remote sensing image of dimensions $H \times W \times 3$, the location feature map T_{loc} is represented by a matrix of size $H \times W$, while the direction feature maps T_{dir} are represented by a tensor of size $H \times W \times (3\text{D}_{\max})$. Here, H and W denote the height and width of the image, respectively, and $D_{\max}$ represents the maximum number of directions associated with each keypoint. Initially, all the keypoints are stored in the binary location mask T^0_{loc} of size $H \times W$, with 1 representing the keypoint. Then, each keypoint in T^0_{loc} is replaced by a Gaussian kernel of size 7×7. As illustrated in Fig. 4, after processing all the keypoints, we obtain the location feature map T_{loc}, in which the element $p_{xy} \in [0, 1]$ represents the validity of the associated keypoint located at (x, y). Subsequently, we construct the direction feature maps T_{dir} by merging T^0_{loc} with the edges determined by the Ramer–Douglas–Peucker algorithm. In detail, for each keypoint in

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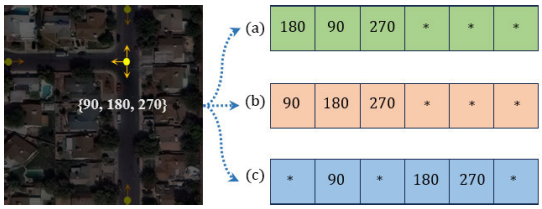


Fig. 5. Diagram of different strategies for direction storage ($D_{\max} = 6$). The number denotes the angle, and the asterisk indicates redundant directions. (a) Directions stored disorderly with redundant directions allocated in the last groups. (b) Directions stored orderly with redundant directions allocated in the last groups. (c) Proposed strategy.

T^0_{loc}, the maximum number of directions $D_{\max}$ is set to 6, with each direction represented as a three-element group. As shown in Fig. 4, for the ith direction of a keypoint located at (x, y), where $i = 1, 2, \ldots, D_{\max}$, the first element represents the validity of the direction $t_{ixy} \in [0, 1]$, and the remaining two elements are the sine and cosine values of the angle, denoted as m_{ixy} and n_{ixy}, respectively. When the number of directions for a keypoint is less than $D_{\max}$, the three-element groups for the redundant directions are set to 0. To maintain consistency with T_{loc}, we expand each three-element group $\{t_{ixy}, m_{ixy}, n_{ixy}\}$ by the identity kernel of size 7×7. Then, three-channel feature maps are generated for each direction, which are concatenated to form the $(3D_{\max})$-channel direction feature maps T_{dir}. We denote the ground truth feature maps as T_{loc} and T_{dir}, with their predicted counterparts represented by T^*_{loc} and T^*_{dir}. Moreover, p^*_{xy}, t^*_{ixy}, m^*_{ixy}, and n^*_{ixy} are the predicted counterparts of p_{xy}, t_{ixy}, m_{ixy}, and n_{ixy}, respectively.

Multiple strategies can be adopted to set redundant directions to zero in T_{dir}. Fig. 5(a) and (b) illustrates methods where redundant directions are allocated to the last few three-element groups, with other valid directions stored in either a disordered or ordered manner based on their angles. However, these approaches do not ensure accurate direction matching during loss computation, leading to instability in supervised training. To address this issue, as shown in Fig. 5(c), we propose dividing 360 degrees into $D_{\max}$ equal intervals. Specifically, each interval spans from $(i-1) \cdot 360/D_{\max}$ degrees to $i \cdot 360/D_{\max}$, where $i = 1, 2, \ldots, D_{\max}$. Directions are classified into specific groups on the basis of their angles, with empty groups designated as redundant directions. However, conflicts arise when multiple directions must be assigned to the same group. To resolve this issue, we employ the Hungarian algorithm [39] to determine the optimal allocation strategy, which minimizes the angular discrepancy of each direction relative to the center of its assigned group.

B. Phase I: Model Training for Feature Map Prediction

As illustrated in Fig. 6(a), KDGraph features two independent decoder branches: the location branch for predicting the location feature map and the direction branch for predicting the direction feature maps. The location branch functions as a classifier to determine the presence of keypoints, while the direction branch classifies the existence of directions and regresses the associated angles. HRNet48 [30] is utilized as the shared encoder to integrate multiscale information from the remote sensing image.

In the following, we detail the architecture of our decoder branch. Each decoder, as illustrated in Fig. 6(a), is composed of a dilated attention module (DAM), two upsample blocks, and two convolutional layers with kernel sizes of 3×3 and 1×1. The convolution operation inherently focuses on local neighborhoods, which limits the network's capacity to capture long-range spatial dependencies and renders it vulnerable to issues such as shadows and occlusions. To mitigate this limitation, we introduce the DAM, designed to effectively capture the extensive spatial context. The DAM architecture, as shown in Fig. 6(b), includes a dilation block, four convolutional block attention modules (CBAMs) [40], and nine residual blocks (ResBlocks). The primary function of the dilation block is to expand the receptive field. As depicted in Fig. 6(c), it features three parallel branches, each utilizing dilated convolutions with varying rates arranged in a cascade configuration, and includes a skip connection between the input and output features. The CBAM effectively implements attention mechanisms along both channel and spatial dimensions, enhancing meaningful features while suppressing less relevant ones. As illustrated in Fig. 7, the input features go through the channel and spatial attention modules to generate attention maps, which are then multiplied with the input for refined feature representation. The residual block consists of two sequential 3×3 convolutions, with a skip connection that links the input directly to the output, facilitating the learning of residuals and enhancing the propagation of features. Within the DAM, a sequence comprising a residual block, a CBAM, and another residual block is applied four times. The upsample block consists of a transpose convolution and three residual blocks. Specifically, the two upsample blocks increase the size of feature maps from $(H/4) \times (W/4)$ to $H \times W$. Following this operation, a 3×3 convolution is applied, followed by a 1×1 convolution, to adjust the channel number to 1 in the location branch or to 18 in the direction branch. A sigmoid function is then applied on p^*_{xy} and t^*_{ixy}.

We apply the modified focal loss [36] (denoted as $\mathcal{L}_{\text{mfocal}}$) to p^*_{xy} and t^*_{ixy}, and the mean squared error loss (denoted as $\mathcal{L}_{\text{mse}}$) to m^*_{ixy} and n^*_{ixy}. The overall loss function is

$$\mathcal{L}\left(T_{\text{loc}}, T^*_{\text{loc}}, T_{\text{dir}}, T^*_{\text{dir}}\right) = \sum_{i=1}^{D_{\max}} \sum_{x=1}^{W} \sum_{y=1}^{H} \begin{pmatrix} \mathcal{L}_{\text{mfocal}}\left(p_{xy}, p^*_{xy}\right) + \mathcal{L}_{\text{mfocal}}\left(t_{ixy}, t^*_{ixy}\right) \\ +\mathcal{L}_{\text{mse}}\left(\left(m_{ixy}, n_{ixy}\right), \left(m^*_{ixy}, n^*_{ixy}\right)\right) \end{pmatrix} \tag{1}$$

where p_{xy} and p^*_{xy} are derived from T_{loc} and T^*_{loc}; t_{ixy}, m_{ixy}, and n_{ixy} are derived from T_{dir}; and t^*_{ixy}, m^*_{ixy}, and n^*_{ixy} are derived from T^*_{dir}. In practice, we assign high weights to positive samples of p_{xy} and t_{ixy}, and compute losses only for positive samples of m_{ixy} and n_{ixy}.

C. Phase II: Road Graph Inference

In Phase II, we focus on inferring the road graph based on keypoint features predicted by the model trained in Phase I.

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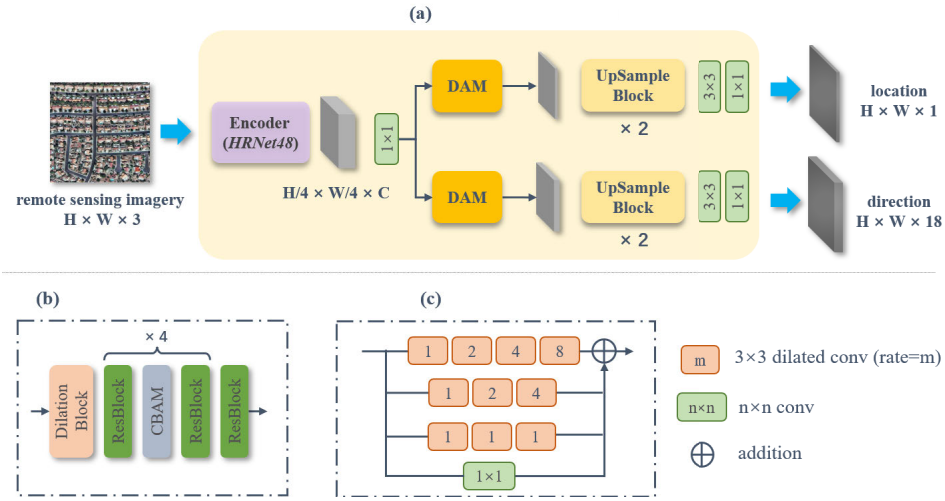


Fig. 6. Network architecture of the proposed KDGraph. (a) Network architecture. (b) DAM. (c) Dilation block.

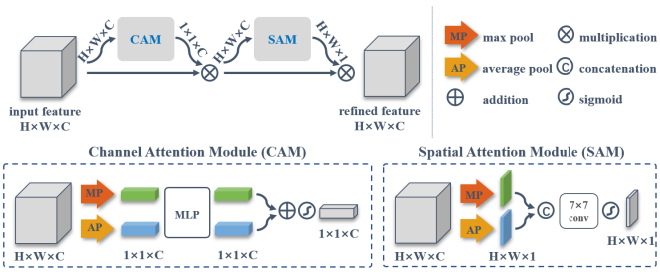
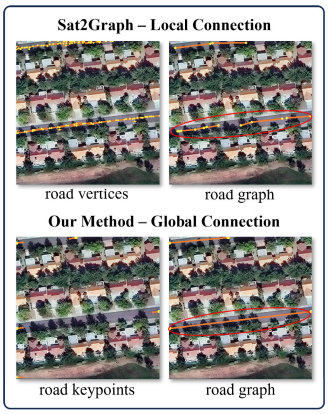
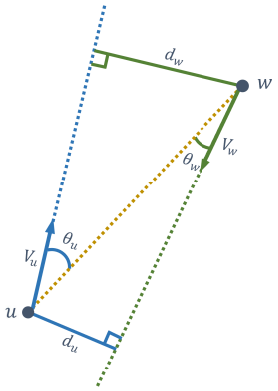


Fig. 7. Architecture of the convolutional block attention module.



(a)



(b)

Fig. 8. Comparison of local and global keypoint connection strategies. (a) Comparison of local and global connection strategies. (b) Global connection strategy.

First, we use the trained model to generate predicted location feature map T^*_{loc} and predicted direction feature maps T^*_{dir}. Subsequently, keypoints are extracted as described in Section II-C1, and these keypoints are connected to form road edges as outlined in Section II-C2.

1) Keypoint Detection From Keypoint Feature Maps: First, we extract keypoint location information from the predicted location feature map T^*_{loc} using a two-stage approach. In stage 1, we sequentially process each pixel in T^*_{loc}. A pixel is selected for further processing if its validity exceeds those of its fixed-size neighbors. In stage 2, the selected pixels with validity greater than a preset threshold τ_{loc} are identified as valid keypoints. Next, we derive the direction information for each valid keypoint from predicted direction feature map T^*_{dir}. For each valid keypoint at coordinate (x, y), the directions in T^*_{dir} are represented as $D_{\max}$ three-element groups $\{t^*_{ixy}, m^*_{ixy}, n^*_{ixy}\}$ with $i = 1, 2, \ldots, D_{\max}$. The values m^*_{ixy} and n^*_{ixy} are normalized to ensure angular validity. Then, we retain the angle whose direction validity t^*_{ixy} exceeds a preset threshold τ_{dir} for the next step. In our experiments, τ_{loc} and τ_{dir} are set to 0.35 and 0.3, respectively. Given that directions of a keypoint are typically not closely clustered, we aggregate nearby directions by removing those with low probabilities. The resulting keypoints, including both locations and directions, are considered candidate keypoints for connection.

2) Greedy Parsing for Keypoint Connection: We propose a greedy parsing algorithm that globally connects candidate keypoints without distance constraints. In contrast, Sat2Graph employs a local connection strategy that links road vertices within a fixed range. As shown in Fig. 8(a), this local strategy fails to connect the detected vertices in the shaded area, leading to fragmented road segments. However, our greedy parsing algorithm successfully connects keypoints at both ends of the road across long distance, even in challenging regions.

The proposed greedy parsing algorithm consists of two stages: coarse edge extraction and fine edge extraction. Coarse edge extraction is utilized to identify all potential edges. For two keypoints, u and w, as illustrated in Fig. 8(b), their respective direction vectors are denoted as V_u and V_w. We compute the sum of the perpendicular distances, $d = d_u + d_w$, from the keypoints to the direction vectors, and the angle between the directions, $\theta = \theta_u + \theta_w$. The potential edge is documented when conditions $d < \tau_d$ and $\theta < \tau_{\text{angle}}$ are met. In this study, τ_d, ranges from 70 to 140, increasing when the distance between keypoints, while τ_{angle} is set to 45. However, multiple

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potential edges may be associated with the same keypoint direction. Fine edge extraction further refines these potential edges to select the most suitable one. Specifically, we define the cost of each potential edge as $(d_u + d_w) * \lambda_{\text{dis}}$, where $\lambda_{\text{dis}} \in [0, 1]$ represents the normalized distance between keypoints. These edges are then sorted by cost in ascending order. Starting with the edge of lowest cost, we incorporate it into the road graph and eliminate other potential edges with repetitive directions. This process continues until all potential edges have been fully processed.

3) Large-Scale Road Graph Inference: In practical applications, large-scale road graph extraction is more common than small-region extraction. We evaluate the performance of different strategies for large-scale road graph inference using space and time complexity. Space complexity refers to the average memory usage during processing, while time complexity represents the time required to generate the road graph.

Edge-first methods partition large images into overlapping patches, which are subsequently stitched together using an averaging operation to generate a road mask. This mask is then transformed into a road graph. However, the time required for graph conversion increases exponentially with image size, due to road loops and variable junction links. Iterative vertex-first methods extract road segments through iterative strategies and merge vertices when they are sufficiently close. The time complexity of this strategy scales rapidly with image size as the number of extracted vertices grows. Sat2Graph, based on an evenly spaced strategy, follows a similar approach to edge-first methods by generating vertex feature maps for large images and connecting vertices to form the road graph. Although it is faster than previous methods, it requires substantial memory to store intermediate feature maps, significantly increasing space complexity.

In this study, we propose a patch expansion strategy for large-scale road graph inference, which considerably reduces the time complexity. In the proposed KDGraph, keypoints can appear at patch boundaries, leading to inconsistencies in the overlapping regions between adjacent patches, which makes averaging operations impractical. To address this issue, each patch must independently extract its section of the road graph. While cropping patches without overlap ensures independence, it fails in complex scenarios. For example, if two adjacent patches split a T-junction into perpendicular road segments, the junction's topology will not be captured due to the absence of keypoints in both patches.

Our patch expansion strategy, as depicted in Fig. 9, involves several key steps. Initially, the large image is segmented into nonoverlapping patches. Each of these patches is then expanded to include an auxiliary region, which provides additional context and information for the initial patch. Subsequently, the road graph is generated within each expanded patch and adjusted to fit the dimensions of the original patch. The final step involves stitching together the road graphs from adjacent patches by merging keypoints located near the intersection lines of neighboring patches. This approach optimizes space complexity by storing only road vertices during processing. Additionally, it significantly reduces time complexity by decomposing the large-scale road graph extraction into small and manageable tasks, with time consumption increasing linearly with the image size.

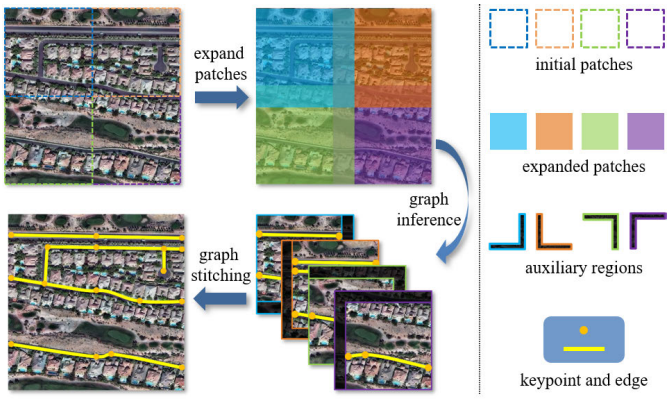


Fig. 9. Processing pipeline of the patch expansion strategy in the proposed method.

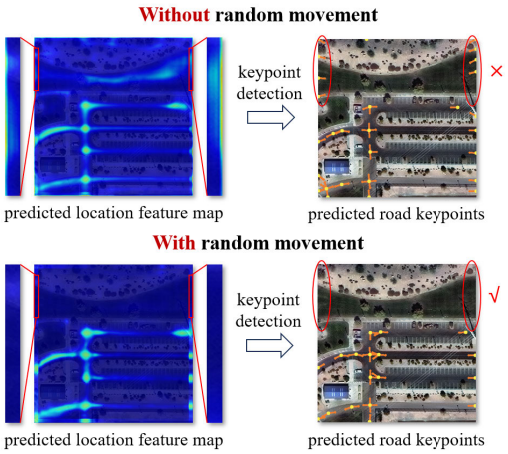


Fig. 10. Comparison of scenarios with and without random movement of keypoints at the image edges during training.

D. Random Movement of Keypoints on the Image Edges

Keypoints are predominantly located on the edges of images. Statistical analysis of the SpaceNet3 dataset [37] reveals that 24.56% of ground truth road keypoints are situated on image edges, despite edge pixels comprising only 0.31% of the total image area. This results in a probability of 0.25% for edge pixels being keypoints, which is 100 times higher than that for pixels located within the image. The significant imbalance in keypoint distribution can potentially mislead the network into erroneously assign high probabilities to image edge points.

To address this issue, we randomly move keypoints located on the image edges inward during training, while preserving their associated directions. The displacement distance is randomly chosen from an integer range from 1 to 10 pixels. Fig. 10 demonstrates that our strategy significantly reduces false keypoint predictions on the image edges during inference. This suggests that the network can be effectively trained to generate accurate feature values along the image edges.

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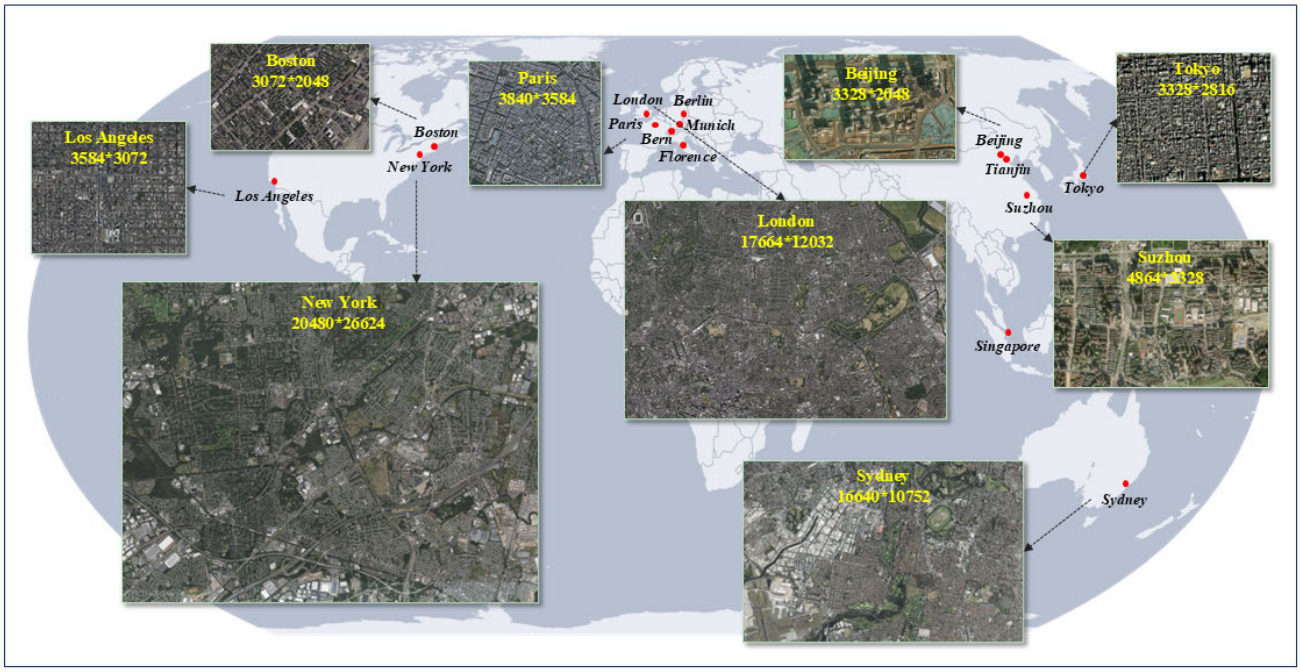


Fig. 11. Data distribution of the SoR dataset.

III. Experiments and Analysis

A. Dataset

1) SpaceNet3: The SpaceNet3 dataset consists of 8000 km of road centerlines annotated in GeoJson format. The images are acquired from the WorldView-3 satellite sensor, covering four regions: Vegas, Paris, Shanghai, and Khartoum. These images are 8-band multispectral with 11-bit radiometric resolution, and each tile represents a 400 × 400 m area with a pixel resolution of 0.3 m. For our experiments, we convert the multispectral images to 8-bit RGB images using min–max normalization and enhance image details through histogram equalization. Since testing data annotations are not available, we utilize 2549 training tiles, which are split into training (80%), testing (15%), and validation (5%) sets. To manage memory usage, training images are cropped into patches of 448 × 448 pixels with a 128-pixel overlap.

2) SoR Dataset: To assess the generalization capability and effectiveness of various methods in handling shadows and occlusions, we have developed the SoR dataset. Satellite images were acquired from Bing Map[1] with a resolution of 0.6 m, while road vector data were sourced from OpenStreetMap (OSM) [41]. As illustrated in Fig. 11, the SoR dataset encompasses satellite images from 15 cities around the world, each exhibiting varying degrees of shadow and occlusion. The specifications for image sizes and coverage areas are exhibited in Table I. As demonstrated in Fig. 12, the satellite images from various regions exhibit distinct data distributions, which facilitates the evaluation of the generalization capability of different methods. Given the imperfections in the OSM road data, we manually correct the inaccuracies within the dataset. All images are employed to evaluate the generalization capabilities of methods trained on the SpaceNet3 dataset.

[1]https://learn.microsoft.com/en-us/bingmaps/rest-services/imagery/get-a-static-map



Fig. 12. Visualization of sample images from the SoR Dataset.

B. Implementation Details

We employ a learning rate of 0.01 to train the model for 200 epochs with the AdamW optimizer. All experiments are conducted on a single NVIDIA GeForce RTX 3090 with 24 GB of GPU memory. Due to memory constraints, we implement a gradient accumulation strategy to effectively increase the batch size to 16. Data augmentation techniques, including random rotation and random flip, are employed. The ReduceLROnPlateau scheduler in Pytorch [42] is utilized,

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TABLE I
REMOTE SENSING DATA OVERVIEW FOR THE SOR DATASET

Region	Area (km^2)	Image size	Region	Area (km^2)	Image size	Region	Area (km^2)	Image size
NewYork	196.29	$26,624 \times 20,480$	Berlin	9.44	$5,120 \times 5,120$	LosAngeles	3.96	$3,584 \times 3,072$
London	76.51	$17,664 \times 12,032$	Munich	3.11	$3,072 \times 2,816$	Boston	2.26	$3,072 \times 2,048$
Sydney	64.41	$16,640 \times 10,752$	Florence	2.55	$3,072 \times 2,304$	Suzhou	5.83	$4,864 \times 3,328$
Beijing	2.45	$3,328 \times 2,048$	Bern	2.76	$3,328 \times 2,304$	Tokyo	3.37	$3,328 \times 2,816$
Paris	4.95	$3,840 \times 3,584$	Tianjin	3.37	$3,328 \times 2,816$	Singapore	1.16	$1,792 \times 1,792$

TABLE II
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON THE SPACENET3 DATASET

Type	Method	Precision	Recall	F1	APLS
Edge-first	HRNet48	0.8346	0.5847	0.6876	0.6671
	GAMSNet	0.8334	0.6000	0.6977	0.6682
	CoANet	0.7905	0.5748	0.6656	0.6598
	DDUNet	0.8313	0.5449	0.6583	0.5965
	MSSDMPA	0.8120	0.5721	0.6712	0.6219
Vertex-first	Sat2Graph	**0.9294**	0.4789	0.6321	0.4970
	RNGDet	0.8326	0.5027	0.6269	0.6498
	KDGraph	0.8371	**0.6165**	**0.7101**	**0.6734**

which reduces the learning rate by a factor of 2 if there is no significant reduction in training loss over three consecutive epochs.

C. Benchmark Evaluation Metrics and Methods

We assess the performance of various methods using two topological metrics: TOPO [43] and APLS [37]. TOPO computes the average precision, recall, and F_1-score for subgraphs sampled from the ground truth and predicted graphs. APLS measures the discrepancy in the lengths of shortest paths between selected node pairs in the two graphs.

For comparison, the following methods are evaluated on both datasets: edge-first methods, including HRNet48 [30], GAMSNet [32], CoANet [33], DDUNet [31], and MSSDMPA [34], and vertex-first methods, including Sat2Graph [6] and RNGDet [14].

D. Experimental Results on the SpaceNet3 Dataset

Table II presents the quantitative results of comparative experiments conducted on the SpaceNet3 dataset, with the best results highlighted in bold. Although vertex-first methods excel in capturing the connection information between road vertices, they do not show significant improvements over edge-first methods. This is attributed to roubustness of edge-first methods in handling typical road conditions. Our method demonstrates superior performance in TOPO recall, TOPO F1, and APLS, highlighting the effectiveness of the novel keypoint detection approach in road graph extraction. Specifically, our method surpasses the second-best method, GAMSNet by 2.75% in TOPO precision, 1.78% in TOPO F1-score, and 0.78% in APLS.

Fig. 13 visualizes the results of various methods on the SpaceNet3 dataset. Fig. 13(a) and (b) illustrates typical cases involving shadows and occlusions caused by tall buildings or trees. Shadows and occlusions can lead to discontinuous road representations in remote sensing images, resulting in incomplete road graphs when using edge-first methods. In contrast, the proposed KDGraph accurately extracts road topology in shadowed or occluded areas, demonstrating the effectiveness and robustness of the novel keypoint detection in such scenarios. Fig. 13(c) displays an area with small roads, which occupy less space than ordinary roads, leading to weaker road representations in remote sensing images. As a result, these roads are often overlooked or incompletely extracted by existing methods. However, similar to its performance in shadowed and occluded areas, KDGraph successfully detects keypoints at both ends of small roads and accurately extracts their topology, due to the clear topological significance of the

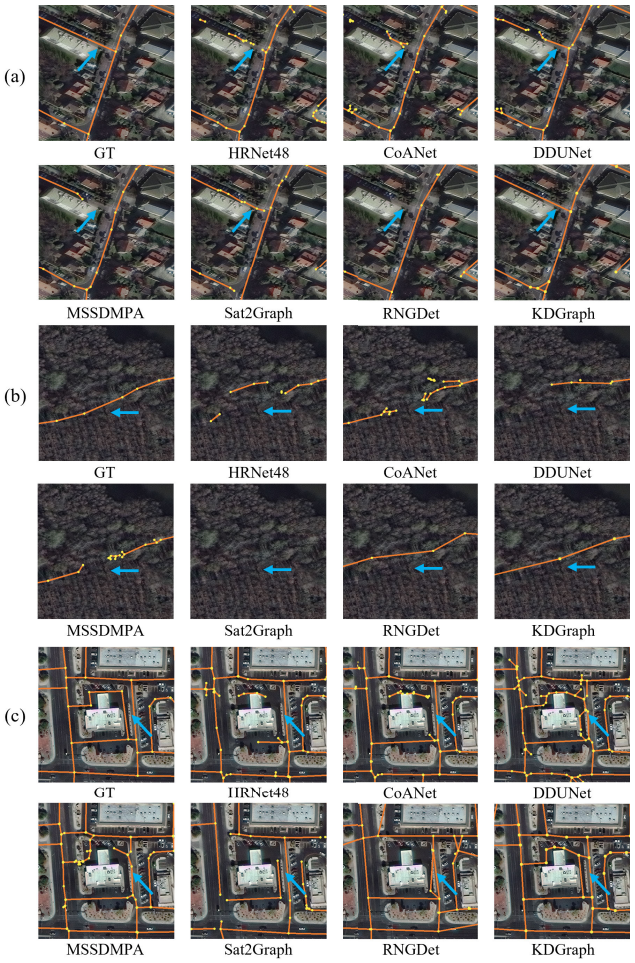


Fig. 13. Visualization of comparative methods on the SpaceNet3 dataset. (a) and (b) Illustrates cases involving shadows and occlusions caused by tall buildings or trees. (c) Typical case with small roads.

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road keypoints. In conclusion, KDGraph effectively mitigates the negative impacts of shadows and occlusions and can handle more complicated scenarios than existing methods.

E. Experimental Results on the SoR Dataset

We use models trained in the SpaceNet3 dataset to directly perform large-scale road graph extraction on the SoR dataset to evaluate generalizability. Specifically, all experiments use a patch expansion strategy with an input block size of 448 × 448 and a 128-pixel overlap. We ensure consistency and fairness by retaining the same hyperparameters used during training. The only exception is that, for all methods, we recalculate the mean and standard deviation for normalization in each city. In these experiments, no fine-tuning is performed.

Experimental results evaluated using APLS on the SoR dataset are presented in Table III. It is important to note that the computational complexity of APLS increases exponentially with the size of the road graph. Therefore, for New York, London, and Sydney, we partition the road graph into nonoverlapping 3500 × 3500 subgraphs and compute the final APLS score by averaging the results across these subgraphs. KDGraph demonstrates superior performance, achieving the highest APLS in 12 out of 15 cities, with significant improvements observed in most cases. We attribute the strong generalization ability of KDGraph primarily to its novel keypoint detection approach, which reformulates road mask feature learning into keypoint detection. By concentrating on keypoints at typical road line locations, KDGraph simplifies the task compared to road mask extraction or equal-interval road vertex extraction, reducing the impact of varying data distributions across different cities.

The visualization results of GAMSNet, Sat2Graph, and KDGraph in Berlin are shown in Fig. 14. We focus on two regions with shadows and occlusions. GAMSNet and Sat2Graph produce fragmented or incorrect road segments in shadowed areas, whereas KDGraph accurately captures the road topology, benefiting from its novel keypoint detection method. Fig. 15 displays visualization results for Sydney, with three regions highlighted in red, blue, and green. The red and blue regions represent typical scenarios with border trees along the road. GAMSNet and Sat2Graph exhibit limitations in accurately extracting road topology in occluded areas, whereas KDGraph effectively detects keypoints and mitigates the effects of occlusions. The green region highlights a complex road junction, common in urban areas. GAMSNet and Sat2Graph fail to accurately extract the junction topology due to limitations in their road graph conversion algorithms and disorganized road vertices, respectively. In contrast, KDGraph, leveraging the clear topological significance of keypoints, accurately captures the junction topology, which demonstrates its effectiveness in handling complex urban scenarios.

To demonstrate the effectiveness of the patch expansion strategy for large-scale road graph extraction, we conduct experiments on Berlin and Sydney with and without patch expansion. The results are exhibited in Table IV. In Berlin, edge-first methods and RNGDet exhibit significantly longer processing times compared to Sat2Graph and KDGraph. In Sydney, the time consumption of these methods grows exponentially with the image size. While Sat2Graph is fast, its memory usage for intermediate feature maps increases rapidly with larger images, leading to out-of-memory (OOM) issues. Conversely, the time consumption of edge-first methods and RNGDet decreases significantly with patch expansion

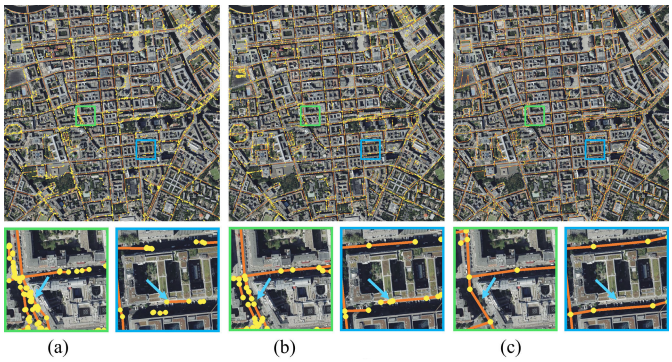


Fig. 14. Visualization results of GAMSNet, Sat2Graph, and KDGraph in Berlin (image size: 5120 × 5120 pixels). (a) GAMSNet. (b) Sat2Graph. (c) KDGraph.

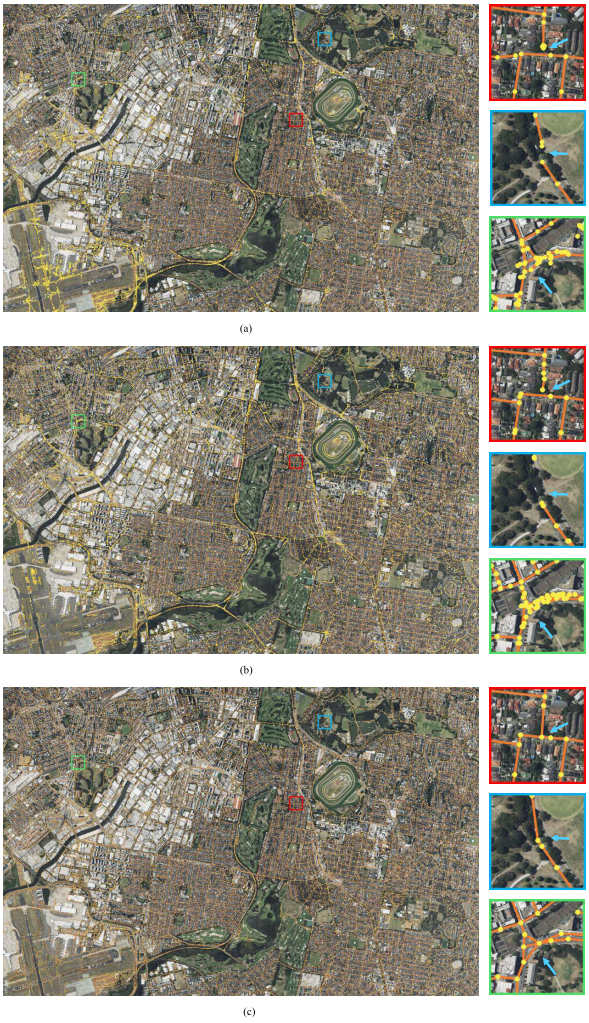


Fig. 15. Visualization results of GAMSNet, Sat2Graph, and KDGraph in Sydney (image size: 16 640 × 10 752 pixels). (a) GAMSNet. (b) Sat2Graph. (c) KDGraph.

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TABLE III
EXPERIMENTAL RESULTS EVALUATED USING APLS ON THE SOR DATASET. NY REPRESENTS NEW YORK, AND LA REPRESENTS LOS ANGELES

Method	NY	LA	Boston	London	Paris	Berlin	Munich	Florence	Bern	Sydney	Beijing	Tianjin	Suzhou	Tokyo	Singapore
HRNet48	0.3192	0.6955	0.1391	0.3089	0.1597	0.3767	0.3805	0.2161	0.0202	0.3301	0.0155	0.1229	0.0366	0.1434	0.2519
GAMSNet	0.3661	0.7476	0.2533	0.3826	**0.3124**	**0.3799**	0.5416	0.4384	0.0351	0.3983	0.0301	0.1411	0.0454	0.1188	0.3190
CoANet	0.2550	0.6612	0.2873	0.3030	0.2094	0.2168	0.1750	0.1221	0.0091	0.3705	0.0264	0.0148	0.0195	0.0269	0.0649
DDUNet	0.2470	0.7377	0.1426	0.2293	0.0502	0.2003	0.1493	0.0881	0.0229	0.2984	0.0094	0.0301	0.0325	0.0114	0.2520
MSSDMPA	0.2284	0.6024	0.2012	0.2391	0.0432	0.2843	0.2675	0.0832	0.0231	0.2719	0.0231	0.0895	0.0246	0.0932	0.2814
Sat2Graph	0.1624	0.4299	0.0995	0.1339	0.0221	0.0133	0.1755	0.1250	0.0162	0.2129	0.0271	0.1054	0.0246	0.0385	0.3055
RNGDet	0.3171	0.7252	0.2135	0.3619	0.1475	0.0538	0.1298	0.1533	**0.0812**	0.3231	0.0239	0.0328	0.0531	0.1213	0.2314
KDGraph	**0.4119**	**0.7774**	**0.3882**	**0.4253**	0.1433	0.3554	**0.6760**	**0.5295**	0.0561	**0.4662**	**0.0528**	**0.1809**	**0.0785**	**0.2494**	**0.5947**

TABLE IV
EXPERIMENTAL RESULTS EVALUATED USING TIME AND SPACE COMPLEXITY IN BERLIN AND SYDNEY ON THE SOR DATASET. PE DENOTES PATCH EXPANSION; OOM INDICATES OOM. TIME COMPLEXITY (HOUR) IS LISTED BEFORE THE SLASH, AND SPACE COMPLEXITY (GB) AFTER

Method	Berlin		Sydney	
	w/o PE	w/ PE	w/o PE	w/ PE
HRNet48	0.75 / 3.12	0.05 / 2.87	85 / 5.54	0.44 / 2.87
GAMSNet	5.50 / 4.35	0.14 / 4.15	177 / 6.81	0.82 / 4.15
CoANet	0.87 / 3.20	0.04 / 2.66	94 / 6.84	0.46 / **2.66**
DDUNet	0.85 / 21.20	0.05 / 4.80	95 / 7.45	0.45 / 4.80
MSSDMPA	5.80 / 5.13	0.22 / 4.93	181 / 6.27	1.00 / 4.93
Sat2Graph	0.07 / 5.86	0.06 / 3.91	OOM	0.42 / 3.91
RNGDet	1.80 / 2.58	0.14 / **2.52**	102 / 2.82	0.93 / 2.69
KDGraph	-	**0.02** / 3.08	-	**0.28** / 4.83

TABLE V
ABLATION STUDIES OF THE ARCHITECTURE OF DAM ON SPACENET3

+ Dilation block	+ CBAM	F1	APLS
-	-	0.6958	0.6451
✓	-	0.6942	0.6532
-	✓	0.6965	0.6547
✓	✓	**0.7101**	**0.6734**

TABLE VI
ABLATION STUDIES OF THE BACKBONE ON SPACENET3

Backbone	F1	APLS
Resnet50	0.6591	0.6124
Resnet101	0.6825	0.6504
Res2net101	0.6739	0.6308
Swin-S	0.6134	0.5676
Swin-B	0.5150	0.4642
HRNet32	0.6810	0.6423
HRNet48	**0.7101**	**0.6734**

TABLE VII
ABLATION STUDIES OF THE DECODER STYLE ON SPACENET3

Decoder Style	F1	APLS
Integrated	0.6345	0.5923
Independent	**0.7101**	**0.6734**

TABLE VIII
ABLATION STUDIES OF DIRECTION STORAGE STRATEGIES ON SPACENET3

Strategy	F1	APLS
Without order	0.2784	0.0518
With order	0.6523	0.6050
Our strategy	**0.7101**	**0.6734**

TABLE IX
ABLATION STUDIES OF τ_{loc} AND τ_{dir} ON SPACENET3

τ_{loc}	τ_{dir}	Precision	Recall	F1	APLS
0.3	0.25	0.8303	**0.6192**	0.7094	0.6661
0.3	0.3	0.8397	0.6190	**0.7127**	0.6658
0.35	0.25	0.8295	0.6167	0.7074	0.6727
0.35	0.35	**0.8427**	0.6124	0.7093	0.6670
0.35	0.3	0.8371	0.6165	0.7101	**0.6734**

TABLE X
ABLATION STUDIES OF τ_d AND τ_{angle} ON SPACENET3

τ_d	τ_{angle}	Precision	Recall	F1	APLS
70-140	30	**0.8511**	0.5841	0.6928	0.6601
70-140	60	0.7742	**0.6631**	**0.7144**	0.6591
70-120	45	0.8421	0.6034	0.7030	0.6693
50-140	45	0.8401	0.6021	0.7015	0.6659
70-140	45	0.8371	0.6165	0.7101	**0.6734**

TABLE XI
CONFLICT RATES AND KEEP RATES OF DIFFERENT $D_{\max}$ ON SPACENET3

D_{max}	3	4	5	6	7
Conflict rate	8.76%	3.69%	0.93%	0.57%	0.41%
Keep rate	97.61%	99.90%	99.99%	100%	100%

and scales linearly with image size. With patch expansion, Sat2Graph successfully extracts the road graph in Sydney on a single GPU. The memory usage for most methods decreases because intermediate feature maps of large images are no longer required. Notably, KDGraph consistently maintains superior time complexity across all experiments.

F. Visualization Analysis

We visualize the intermediate feature outputs of GAMSNet, Sat2Graph, and KDGraph in Fig. 16. Fig. 16(I) and (II) illustrates scenarios with vegetation and shadows on the road, respectively. The weak road representation caused by vegetation or shadows results in diminished road features in the feature maps of GAMSNet and Sat2Graph, leading to incomplete road topology in these areas. In contrast, the

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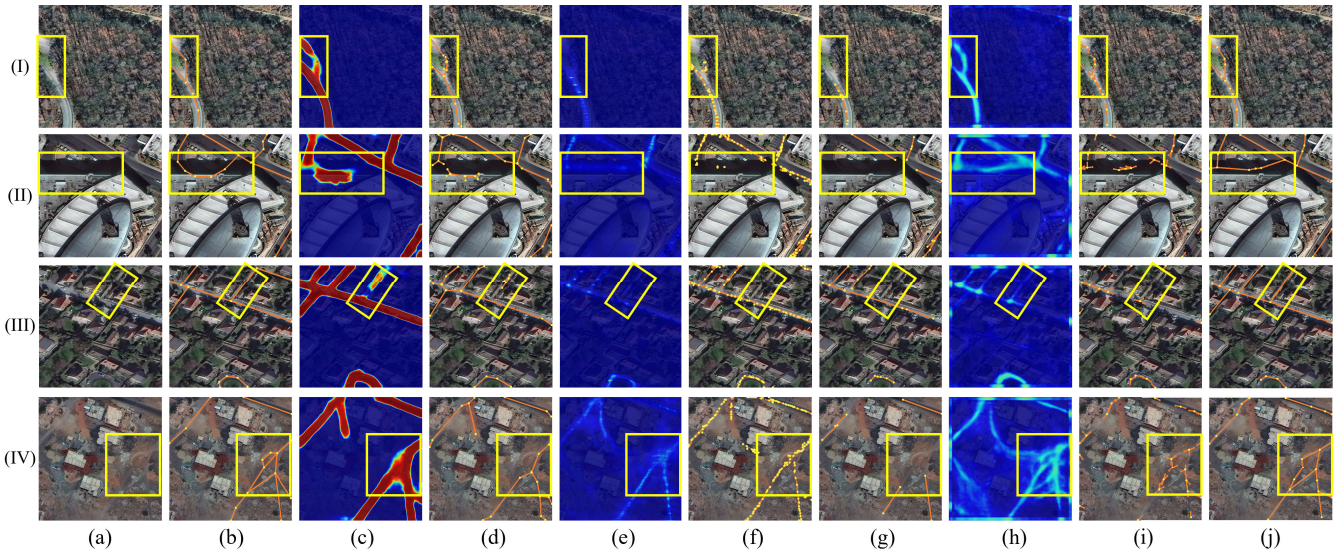


Fig. 16. Visualization results of GAMSNet and KDGraph under diverse conditions. (a) Original image, (b) ground truth, (c) predicted heatmaps of GAMSNet, (d) results of GAMSNet, (e) location features of Sat2Graph, (f) predicted candidate vertices of Sat2Graph and (g) results of Sat2Graph, (h) location feature maps of KDGraph, (i) predicted candidate keypoints of KDGraph, and (j) results of KDGraph.

location feature map of KDGraph retains strong keypoint representations in occluded or shaded areas, demonstrating the robustness of the novel keypoint detection method against shadows and occlusions in remote sensing images. Fig. 16(III) presents visualizations of small roads. The weak representation of small roads leads to low feature values using GAMSNet and Sat2Graph, causing false negative topology in these areas. However, KDGraph successfully extracts strong keypoint representations at both ends of small roads, demonstrating that road keypoints are more sensitive to road topology than vertices in Sat2Graph. We present the comparison results in rural regions in Fig. 16(IV). In contrast to urban areas characterized by well-defined driveways, rural regions exhibit complex roads composed of diverse materials, which substantially increase the complexity of road graph extraction. Despite the challenges in distinguishing road lines in rural areas, KDGraph's location feature map retains high activation values. As a result, KDGraph produces a more accurate road topology than GAMSNet and Sat2Graph, attributable to the precise topological information encoded by road keypoints. In conclusion, the proposed keypoint detection method demonstrates strong robustness and effectiveness in handling shadows, occlusions, and other complex scenarios, ensuring more accurate road graph extraction in challenging conditions than existing methods.

G. Ablation and Discussion

1) Architecture of DAM: The proposed DAM is composed of a dilation block, four CBAMs, and nine residual blocks. The dilation block enhances the receptive field by integrating multiscale features through parallel branches, which improves KDGraph's robustness to shadows and occlusions. The CBAM selectively enhances relevant features in both the location and direction branches while suppressing irrelevant ones, thereby improving keypoint extraction accuracy and facilitating keypoint connection. The residual block consists of two convolutional layers with a residual connection, which facilitates effective learning and deep integration of features. Given its straightforward design, no ablation studies are conducted on the residual block. The results of the ablation study on the DAM architecture, presented in Table V, reveal that removing either the dilation block or CBAM significantly degrades performance. Additionally, we provide visualizations of the feature maps extracted by models with different DAM configurations in Fig. 17. Fig. 17(a)–(d) depicts a scenario with large-area shadows. In the absence of CBAM, both the location and direction feature maps exhibit excessive high features at irrelevant locations, leading to false-positive road segments in the extracted road graph. Moreover, removing the dilation block results in the failure to detect keypoints at the image edges due to the reduced receptive field. Fig. 17(e)–(h) shows a tree-occluded area, where the removal of either the dilation block or CBAM introduces significant inaccuracies in keypoint predictions, causing severe errors in the subsequent keypoint connection process.

2) Backbone Choices: Table VI compares the performance of different backbones, including ResNet50 [44], ResNet101, Res2Net101 [45], HRNet32, HRNet48, and the small and base versions of Swin Transformer [46]. The results indicate that HRNet48 outperforms all other backbones across all evaluation metrics. The pretrained Swin Transformer models, optimized for 224×224 input images, experience notable performance degradation when adapted to the 448×448 input size in our experiments. In contrast, HRNet48 demonstrates a strong capacity to preserve high-resolution representations throughout the encoding process, which is a crucial factor for accurate keypoint detection.

3) Decoder Style: We investigate the impact of independent decoder design for generating keypoint feature maps in Table VII. In comparison to our independent decoder style, we also evaluate an integrated style, where the location and direction feature maps are concatenated, with a single decoder

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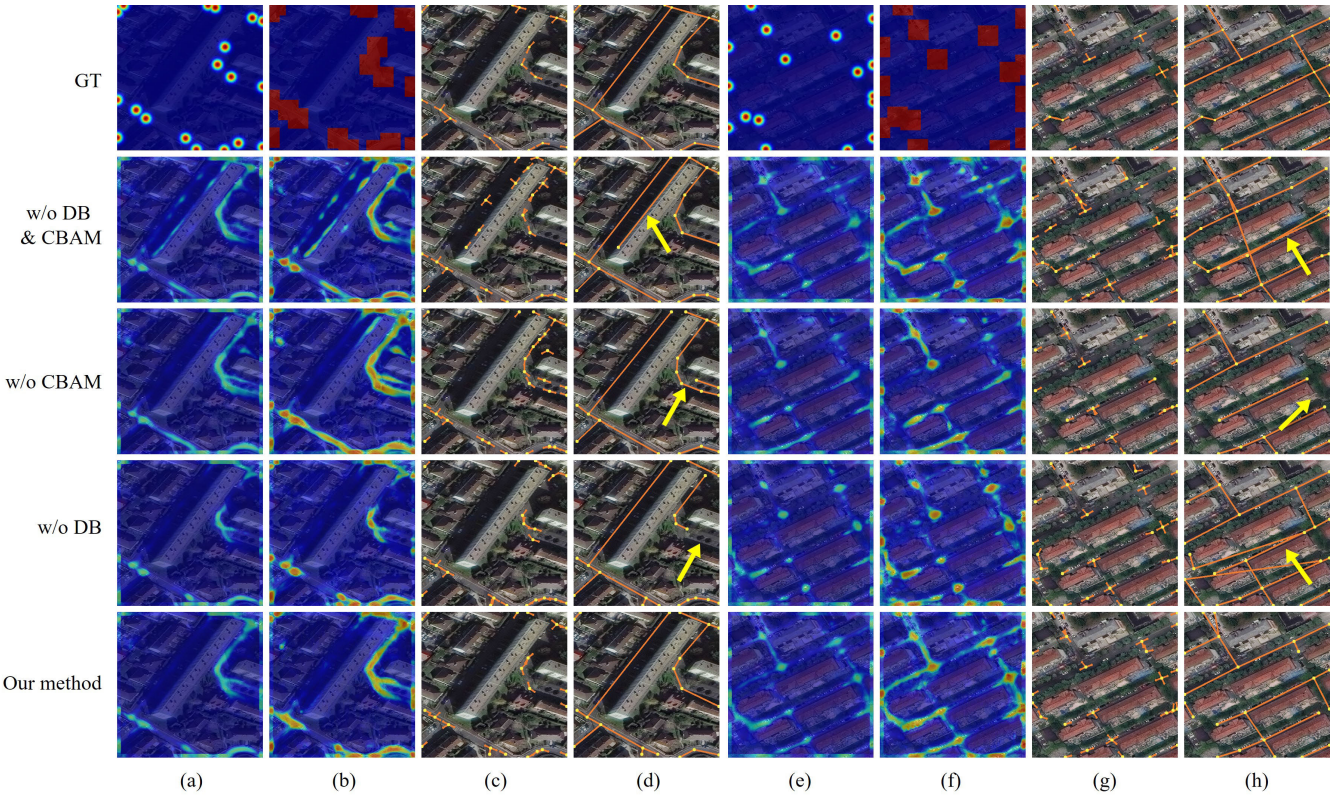


Fig. 17. Visualization of ablation study results for the DAM architecture, where DB denotes the dilation block. (a) and (e) Location feature maps, (b) and (f) maximum probabilities in direction feature maps, (c) and (g) extracted keypoints, and (d) and (h) final road graphs.

responsible for generating the integrated feature maps. Since these two maps capture distinct information, they focus on different encoder features. The results in Table VII show a significant decrease in TOPO F1 and APLS for the integrated style, indicating that a single decoder cannot effectively balance the generation of both feature maps. Therefore, the independent decoder style is selected for generating keypoint feature maps.

4) Strategies for Direction Storage: Table VIII presents the results of ablation studies on different direction storage strategies for generating keypoint feature maps. Storing directions disorderly and placing redundant directions in the last groups (denoted as *without order*) leads to substantial performance degradation for KDGraph due to unstable direction matching during training. Additionally, the orderly storage strategy (denoted as *with order*) results in a lower APLS compared to our approach, as it does not ensure accurate matching between predicted and ground truth directions if predictions are missing or redundant. Our method, which assigns directions to specific groups based on their angle, mitigates the influence of other directions and achieves the best performance in the ablation studies.

5) Different Thresholds: We analyze the effects of thresholds τ_{loc}, τ_{dir}, τ_d, and τ_{angle} as follows. τ_{loc} and τ_{dir} determine the detection of keypoints. As shown in Table IX, lower values of τ_{loc} and τ_{dir} allow more keypoints to be extracted from the feature maps, increasing TOPO recall, whereas higher values improve TOPO precision. τ_d and τ_{angle} affect the number of potential edges in keypoint connection. As shown in Table X, lower values of τ_d and τ_{angle} result in fewer detected potential edges, leading to higher TOPO precision. In this study, we select the configuration with the highest APLS, setting τ_{loc}, τ_{dir}, τ_d and τ_{angle} to 0.35, 0.3, 70–140, and 45, respectively.

6) Discussion on D_{max}: We examine the impact of D_{max} in Table XI. The conflict rate measures the ratio of incorrectly classified directions, while the keep rate indicates the ratio of successfully stored directions. The keep rate drops below 100% when the number of valid directions exceeds D_{max} for certain keypoints. As shown in Table XI, the conflict rate rises with D_{max}, while the keep rate decreases. Given that higher D_{max} increases memory usage for direction feature maps, we choose $D_{\text{max}} = 6$, which maintains all valid directions in groups and exhibits a lower conflict rate compared to smaller D_{max} values.

IV. Conclusion

In this work, we propose KDGraph, a novel keypoint detection method for road graph extraction in remote sensing images. Shadows and occlusions frequently occur in remote sensing images, leading to incomplete road graphs with existing methods. To tackle this issue, KDGraph converts road graph extraction into a keypoint detection task, focusing on vertices at road endpoints, corners, or junctions. Additionally, a greedy parsing algorithm is employed to connect keypoints based on their directions. To assess the generalization and the capability in handling shadows and occlusions of various methods, we constructed a SoR dataset using Bing Map API

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and OSM. We design a patch expansion strategy for large-scale road graph inference. Extensive experiments on both the SpaceNet3 and SoR datasets reveal that KDGraph surpasses other methods in APLS, particularly excelling in shadowed and occluded regions, and demonstrates superior generalization ability. Nonetheless, the current greedy parsing algorithm, which does not employ a deep learning-based approach for keypoint connection, requires adjustment of hyperparameters when spatial resolution changes. Additionally, the patch expansion strategy may result in false connections at the intersection of two adjacent patches. Future research will focus on addressing these limitations.

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